

Mathematical Thinking & Derivation

For many of you, EE 66 might be the first time¹ you are asked to “prove” an idea or a concept. This note tries to explain the main ideas behind proofs and give you some tips on how to approach them.

A mathematical proof provides a means for guaranteeing that a statement is true. So what is a proof? A proof is a finite sequence of steps, called logical deductions, which establishes the truth of a desired statement. In particular, the power of a proof lies in the fact that using limited (finite) means, we can guarantee the truth of a statement with infinitely many cases.

More specifically, a proof is typically structured as follows. Recall that there are certain statements, called axioms or postulates, that we accept without proof (we have to start somewhere). Starting from these axioms, a proof consists of a sequence of logical deductions: Simple steps that apply the rules of logic. This results in a sequence of statements where each successive statement is necessarily true if the previous statements were true. This property is enforced by the rules of logic: Each statement follows from the previous statements. These rules of logic are a formal distillation of laws that were thought to underlie human thinking.

In this note, we are going to guide you through the process of developing proofs with a few examples. In particular, we aim to demonstrate the thought process of turning the problem statement into mathematical form and deriving successive mathematically rigorous statements that leads to the desired result.

When we encounter a proof problem, we generally try to understand the problem by asking the following questions:

- "What are the things we can assume based on the problem statement?"
- "What is it that we would like to show?"

The answer to the first question gives us the conditions we are working under and the answer to the second question gives us a clear picture of our goal. Then, we ask the question:

- "How can we utilize what we know to get to what we would like to show under the specified condition?"

¹CS70 formally teaches proofs to Berkeley EECS and CS undergraduates. Check out their website for good introductions to proofs and mathematical notation.

To write a proof, the following steps are useful in guiding your thought process and helping you understand the fundamental ideas of the problem.

1. **Carefully read the statement.** Check to see what the direction of the implication is: are you being asked to assume P is true and prove Q is true ($P \implies Q$)? Or are you being asked to assume Q is true and prove P is true?
2. **Write out what you know from the statement of the theorem.** Say you are asked to prove something like “If P is true, then Q is true.” Here, “ P is true” is what is given to you/what is known. If the theorem statement has this written out in words, write it out in mathematical notation. Be explicit. Try to simplify any complex notation or jargon.
3. **Write out what you want to prove.** In the example above, what you want to prove is “ Q is true”. Again, this might be given to you in words, make sure you translate it into equations. Simplify. Make sure you write this on the paper, even though it seems trivial, it can often help. If there are different ways of writing this – write out both. For example, if you want to prove a set of vectors is linearly dependent, you have two definitions of linear dependence you can work with. Write out both.
4. **Observe the two statements you just wrote down for what is known and what you want to prove. Find similarities.** Take note of these. It might help to also write them down. How might one of the expressions you wrote be made to look like the other?
5. **Try out a simple example.** Now that you have written out what is given and what you want to prove, try thinking about a simpler version. For example, if you are asked to prove something for an $n \times n$ matrix, try writing out the same statements for a 2×2 matrix. Check again for similarities between what is known and what you want to prove. See if you can prove the theorem for the simpler case of 2×2 .

In the process of trying out an example, you may notice a pattern in your working that extends to the general proof. Don’t be afraid to try several examples, if just one isn’t giving you enough intuition. Trying out examples is also a great way of ensuring that you understand what you want to prove in concrete terms, rather than just as an abstract claim. Consequently, the steps needed to get from “ P is true” to “ Q is true” become clearer and you have a better understanding of what is needed at each step to prove the statement.

6. **Manipulate both sides of the claim. JUSTIFY each step.** After coming up with an algebraic representation of the desired claim, try to manipulate both what you are *given* and what you are *trying* to prove, to see if you can simplify the desired claim. Often, it helps to get rid of complex notation as part of this process - for instance, if you see a summation expressed using $\sum$ notation, it might help to write it out explicitly to get a better understanding of what exactly is being summed.

Still, at this stage it is important to ensure that your manipulations are valid - it’s no use making an amazing simplification if it turns out to be wrong! It can never hurt to break down complex steps into multiple smaller ones, and always ask yourself what conditions are needed for each step to be true. For instance, if you are dividing two quantities, then you should ensure that you are not dividing by zero!

7. **Different approaches.** There are different ways to prove a statement. You might encounter some of the following types of proofs.

- *Direct proofs.* This style of proof directly shows what is to be proven from what is known by doing a series of mathematical and logical steps.
- *Constructive proofs.* Essentially, when you are asked to prove that a certain object “exists”, you can prove the statement by explicitly constructing the object.
- *Proof by contradiction.* Another common technique when asked to prove a claim is to try and show that it is impossible for the claim to *not* be true. This technique tends to be useful when the negation of the desired claim is easier to express algebraically than the claim itself.

No one approach is better or more powerful than any other approach — there is no hierarchy. When you are thinking about proofs try out different approaches to see which one might work.

Now, we’ll look at four examples: one of a direct proof, two constructive proofs, and a proof by contradiction.

Example 2.1 (Direct proof): This proof does not follow any particular style of proof — it directly deduces what is to be proved from what is known. **Suppose that $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_1 + \mathbf{v}_2$ are all solutions to the system of linear equations $\mathbf{A}\mathbf{x} = \mathbf{b}$. Prove that $\mathbf{b}$ must be the zero vector.**

Proof: What does it mean for $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_1 + \mathbf{v}_2$ to be the solutions to $\mathbf{A}\mathbf{x} = \mathbf{b}$? It means these vectors must satisfy the following equations:

$$\mathbf{A}\mathbf{v}_1 = \mathbf{b} \tag{1}$$

$$\mathbf{A}\mathbf{v}_2 = \mathbf{b} \tag{2}$$

$$\mathbf{A}(\mathbf{v}_1 + \mathbf{v}_2) = \mathbf{b} \tag{3}$$

Notice that using distributivity of matrix-vector multiplication, equation (3) can be rewritten as

$$\mathbf{A}\mathbf{v}_1 + \mathbf{A}\mathbf{v}_2 = \mathbf{b}. \tag{4}$$

Now from equation (1) and (2), we can substitute $\mathbf{A}\mathbf{v}_1$ and $\mathbf{A}\mathbf{v}_2$ with the vector $\mathbf{b}$, which leads us to

$$\mathbf{b} + \mathbf{b} = \mathbf{b}. \tag{5}$$

Subtracting $\mathbf{b}$ from both sides of the equation above, we have

$$\mathbf{b} = \mathbf{0}. \tag{6}$$

Hence $\mathbf{b}$ is the zero vector, as desired.

Example 2.2 (Explicitly constructing a linear combination): **1. Let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a set of linearly dependent vectors in $\mathbb{R}^n$. Take any matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$. Prove that the set of vectors $\{\mathbf{A}\mathbf{v}_1, \mathbf{A}\mathbf{v}_2, \dots, \mathbf{A}\mathbf{v}_n\}$ is linearly dependent.**

Proof: (1) What do we know? Based on the problem statement, we know that $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a set of linearly dependent vectors. How do we translate this into mathematical form? Recall the set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is linearly dependent if there exists an index i and scalars α_j ’s such that

$$\mathbf{v}_i = \sum_{j \neq i} \alpha_j \mathbf{v}_j. \quad (7)$$

We rewrite this by removing the sigma notation:

$$\mathbf{v}_i = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \cdots + \alpha_n \mathbf{v}_n. \quad (8)$$

(2) What would we like to show? We would like to show that the set of vectors $\{\mathbf{A}\mathbf{v}_1, \mathbf{A}\mathbf{v}_2, \dots, \mathbf{A}\mathbf{v}_n\}$ is linearly dependent. Again using the definition of linear dependence, we can translate it into a mathematical statement – we would like to show that there exist index k and scalars β_l 's such that

$$\mathbf{A}\mathbf{v}_k = \sum_{l \neq k} \beta_l (\mathbf{A}\mathbf{v}_l). \quad (9)$$

(3) Now, how do we use what we know mathematically from (1) to prove the mathematical statement in (2)? We somehow would like to get vectors of the form $\mathbf{A}\mathbf{v}$. How could we do that? Let's multiply both sides of equation (7) by the matrix $\mathbf{A}$:

$$\mathbf{A}\mathbf{v}_i = \mathbf{A} \left(\sum_{j \neq i} \alpha_j \mathbf{v}_j \right). \quad (10)$$

By distributivity of matrix-vector multiplication, we know that

$$\mathbf{A} \left(\sum_{j \neq i} \alpha_j \mathbf{v}_j \right) = \sum_{j \neq i} \mathbf{A}(\alpha_j \mathbf{v}_j) = \sum_{j \neq i} \alpha_j (\mathbf{A}\mathbf{v}_j). \quad (11)$$

Now, we have that

$$\mathbf{A}\mathbf{v}_i = \sum_{j \neq i} \alpha_j (\mathbf{A}\mathbf{v}_j), \quad (12)$$

which is in exactly the mathematical form we would like to show in (9). So what are the values of the β 's we should choose in (9)? We have $\beta_l = \alpha_l$ for all l .

Hence, we have completed our proof. Therefore, we have explicitly found a linear combination of the columns of matrix $\mathbf{A}$ that gives another column of the matrix.

Example 2.3 (Constructive Proof): Prove that $\text{span} \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\} = \mathbb{R}^2$

First, let's figure out exactly what the question is asking us to prove. The span of two vectors is a set of all vectors that can be formed as a linear combination of those two vectors. $\mathbb{R}^2$ is the set of all (Cartesian) vectors with two real components.

We want to show that these two sets are equal, meaning that no element can be in one set but not the other. Thus, we need to show

1. That every vector in the given span is inside $\mathbb{R}^2$, and that
2. Every vector in $\mathbb{R}^2$ is in the given span.

Let's prove each of these statements in order. First, how do we show that every vector in $\text{span}\left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}\right\}$ is contained within $\mathbb{R}^2$? Let's express what we are given algebraically. Consider an arbitrary vector $\mathbf{u} \in \text{span}\left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}\right\}$. By the definition of span and linear combinations, we know that we can write

$$\mathbf{u} = \alpha \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

for some real scalar coefficients α and β . By the rules of vector arithmetic, we can multiply in our scalar coefficients and simplify to obtain

$$\mathbf{u} = \begin{bmatrix} \alpha + \beta \\ \alpha - \beta \end{bmatrix}.$$

Why is this vector in $\mathbb{R}^2$? Well, since α and β are both real numbers, their sums and differences are both also real. And $\mathbf{u}$ clearly has exactly two components. Thus, $\mathbf{u} \in \mathbb{R}^2$, for arbitrary real scalars α and β . Since $\mathbf{u}$ could have been *any* vector in $\text{span}\left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}\right\}$, we have shown that *any* vector in the given span is inside $\mathbb{R}^2$, which is the first thing we wanted to show.

But we're not done yet! All we've done is shown that $\text{span}\left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}\right\} \subseteq \mathbb{R}^2$ - there might still exist a vector outside of the given span that is still within $\mathbb{R}^2$! Let's now try to show that *every* vector in $\mathbb{R}^2$ lies in the given span, by writing any vector in $\mathbb{R}^2$ as a linear combination of the two given vectors.

We'll start with an arbitrary vector $\mathbf{u} \in \mathbb{R}^2$. By definition, $\mathbf{u}$ has two real components, so it can be written as

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

where u_1 and u_2 are both real scalars. We want to know if $\mathbf{u}$ can be written as a linear combination of the two given vectors, and so must ask whether we can find real scalars α and β (that will depend on u_1 and u_2) such that

$$\alpha \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}?$$

Manipulating the above equation, we obtain the linear system

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}.$$

We want to determine whether there will always exist real solutions α and β to the above linear system, no matter what u_1 and u_2 are.

Writing the linear system explicitly, we get the following 2 equations:

$$\alpha + \beta = u_1 \tag{13}$$

$$\alpha - \beta = u_2 \tag{14}$$

Adding the two equations together we get that $2\alpha = u_1 + u_2$, so $\alpha = \frac{u_1 + u_2}{2}$. Similarly, by subtracting the two equations, we get that $2\beta = u_1 - u_2$, so we get that $\beta = \frac{u_1 - u_2}{2}$.

Therefore

$$\alpha = \frac{u_1 + u_2}{2}$$

$$\beta = \frac{u_1 - u_2}{2}.$$

What have we shown? We've shown that for all u_1 and u_2 , we can produce coefficients α and β such that

$$\alpha \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \mathbf{u}.$$

So all $\mathbf{u} \in \mathbb{R}^2$ can be written as a linear combination of our two vectors! Thus,

$$\mathbb{R}^2 \subseteq \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\},$$

which when combined with our earlier result proves that the two sets are equal, as desired! This completes our proof.

Notice that the second half of this proof was a “constructive proof” since, in order to show that the coefficients α and β existed for all $\mathbf{u}$, we were actually able to construct them explicitly!

Example 2.4 (Proof by contradiction): If a theorem statement asks you to prove that something does not exist (e.g. the inverse of a matrix does not exist, or a unique solution does not exist etc.) it is often useful to consider a proof by contradiction. Assume that a unique solution exists, and then show that there is some other contradiction.

The following is a theorem about the set of solutions for a given system of linear equations expressed in matrix-vector form.

Theorem 2.1: If the columns of $\mathbf{A}$ in the system of linear equations $\mathbf{Ax} = \mathbf{b}$ are linearly dependent, then the

system does not have a unique solution.

Let's walk through this proof step by step: we'll start by assuming we have a matrix $\mathbf{A}$ with linearly dependent columns, and then we will show that this means that the system does not have a unique solution.

Since we are interested in the columns of $\mathbf{A}$, let's start by explicitly defining the columns of $\mathbf{A}$:

$$\mathbf{A} = \begin{bmatrix} | & | & & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \dots & \mathbf{a}_n \\ | & | & & | \end{bmatrix},$$

Now, first, let us write out what is known to us.

By the definition of linear dependence, there exist scalars $v_1, \dots, v_n$ such that $v_1 \mathbf{a}_1 + \dots + v_n \mathbf{a}_n = \mathbf{0}$ where not all of the v_i 's are zero. We can put these v_i 's in a vector

$$\mathbf{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$$

and by the definition of matrix-vector multiplication, we can compactly write the expression above:

$$\mathbf{A}\mathbf{v} = \mathbf{0}$$

where $\mathbf{v} \neq \mathbf{0}$.

What do we want to prove?

We are trying to show that the system of equations $\mathbf{A}\mathbf{x} = \mathbf{b}$ does not have a unique solution.

We know that systems of equations can have either zero, one, or infinite solutions. If our system of equations has zero solutions, then it cannot have a unique solution, so we don't need to consider this case.

Now, let us assume we have a unique solution and arrive at a contradiction.

Now let's consider the case where we have at least one solution, $\mathbf{x}_0$:

$$\mathbf{A}\mathbf{x}_0 = \mathbf{b}$$

$$\mathbf{A}\mathbf{x}_0 + \mathbf{0} = \mathbf{b}$$

$$\mathbf{A}\mathbf{x}_0 + \mathbf{A}\mathbf{v} = \mathbf{b}$$

$$\mathbf{A}(\mathbf{x}_0 + \mathbf{v}) = \mathbf{b}$$

Therefore, $\mathbf{x}_0 + \mathbf{v}$ is also a solution to the system of equations! To complete our proof, we now must check that these two solutions are in fact not equal to each other.

Since $\mathbf{v} \neq \mathbf{0}$, we know that $\mathbf{x}_0 \neq \mathbf{x}_0 + \mathbf{v}$. Therefore, our system has two distinct solutions. We've now proven the theorem.

Note that we can add any multiple of $\mathbf{v}$ to $\mathbf{x}$ and it will still be a solution – therefore, if there is at least one solution to the system and the columns of $\mathbf{A}$ are linearly dependent, then there are infinite solutions.